Prerequisites for Convex Optimization

Maxima and Minima

Maxima and Minima

Definitions

- A **local maximum** of a function f is a point $x = c$ where $f(c)$ is greater than all nearby values:

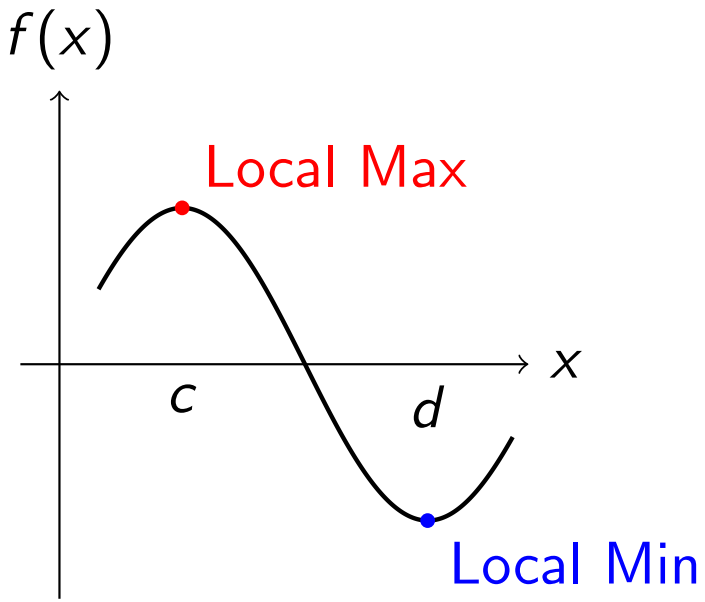
$$f(c) \geq f(x) \quad \text{for all } x \text{ near } c$$

- A **local minimum** of f is a point $x = d$ where $f(d)$ is less than all nearby values:

$$f(d) \leq f(x) \quad \text{for all } x \text{ near } d$$

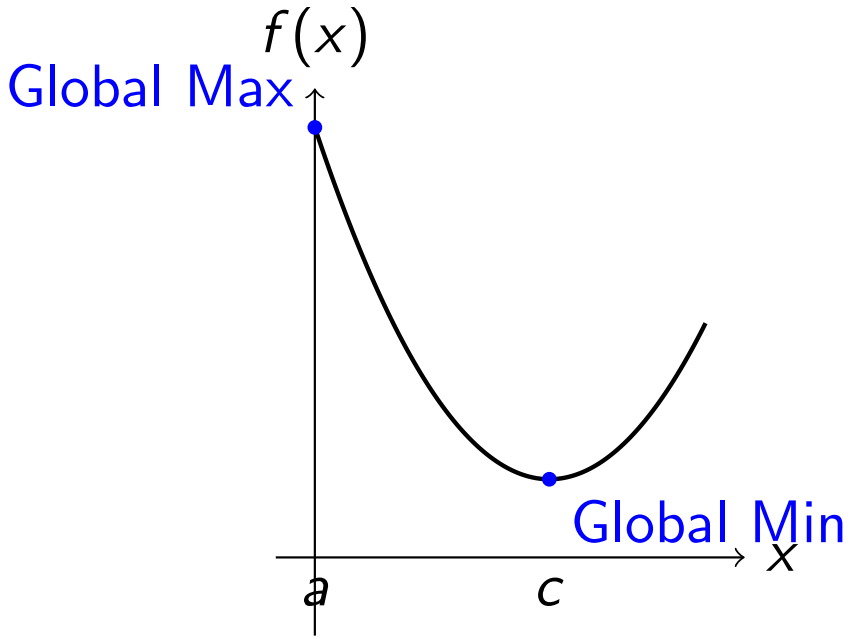
- A **global (absolute) maximum** is the highest value on the entire domain.
- A **global (absolute) minimum** is the lowest value on the entire domain.

Local Maxima and Minima



- Local maxima are points where $f(x)$ changes from increasing to decreasing.
- Local minima are points where $f(x)$ changes from decreasing to increasing.

Global Maxima and Minima



- Global extrema are the absolute highest or lowest values on the entire domain.
- They may occur at local extrema or at domain boundaries.

Methods to Compute Maxima and Minima

Critical Points

- Maxima and minima occur where the first derivative is zero or undefined:

$$f'(x) = 0 \quad \text{or} \quad f'(x) \text{ does not exist}$$

- Points satisfying this are called **critical points**.

First Derivative Test

Procedure

- Find critical points by solving $f'(x) = 0$.
- Analyze the sign of $f'(x)$ around the critical points:
  - If $f'(x)$ changes from positive to negative, f has a local maximum.
  - If $f'(x)$ changes from negative to positive, f has a local minimum.
  - If no sign change, then no local extremum at that point.

Saddle point

A saddle point is a point on the surface of a function where the slope (all derivatives in orthogonal directions) is zero, making it a critical point, but the point is not a local maximum or a local minimum.

Second Derivative Test

Procedure

- Find critical points by solving $f'(x) = 0$.
- Evaluate the second derivative $f''(x)$ at each critical point:
  - If $f''(x) > 0$, the function has a local minimum.
  - If $f''(x) < 0$, the function has a local maximum.
  - If $f''(x) = 0$, the test is inconclusive.

Example: Find Extrema of $f(x) = x^3 - 3x^2 + 4$

Step 1: First derivative

$$f'(x) = 3x^2 - 6x$$

Step 2: Critical points

Solve $f'(x) = 0$:

$$3x^2 - 6x = 0 \implies 3x(x-2) = 0 \implies x = 0,\ x = 2$$

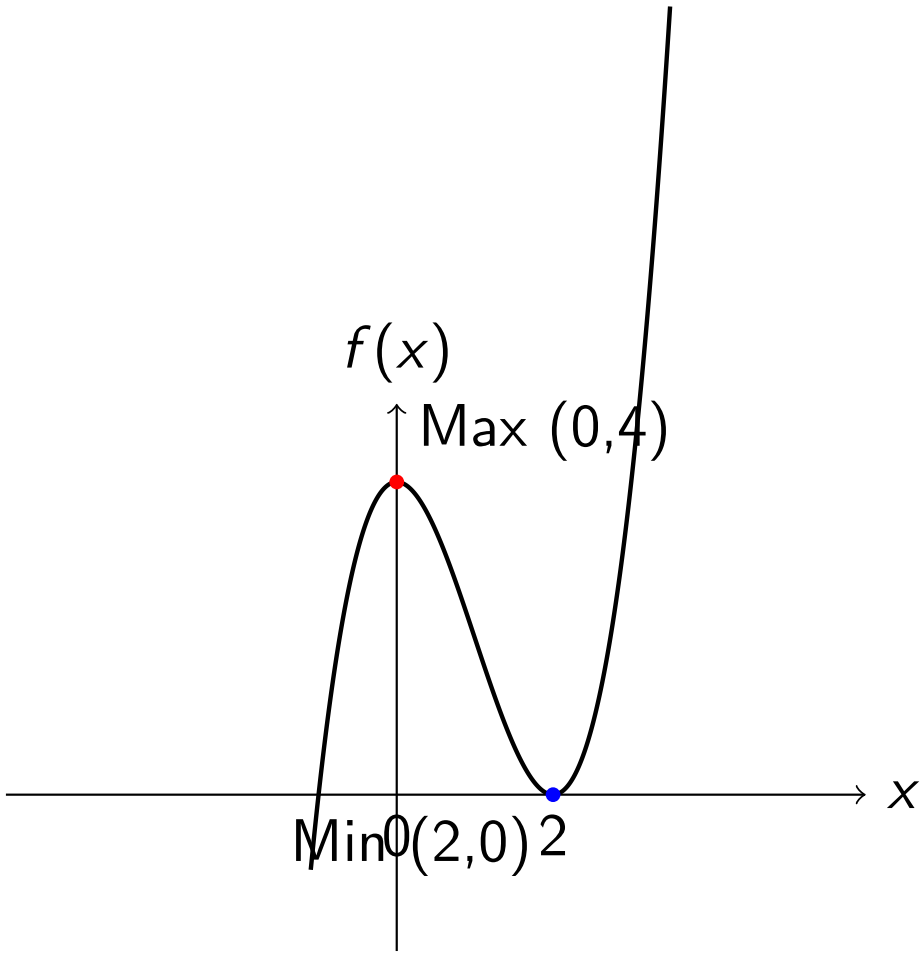
Step 3: Second derivative

$$f''(x) = 6x - 6$$

$$f''(0) = -6 < 0 \implies \text{local maximum at } x = 0$$

$$f''(2) = 6 > 0 \implies \text{local minimum at } x = 2$$

Graphical Representation of Extrema



- The red dot marks the local maximum at $x = 0$.
- The blue dot marks the local minimum at $x = 2$.

Questions: Critical Points, Maxima and Minima

1. Find the critical points of the function $f(x) = x^3 - 3x^2 + 4$.
2. Determine the nature (local max, local min, or saddle point) of the critical points of
$$f(x) = x^4 - 4x^3 + 6x^2.$$
3. For the function $f(x, y) = x^2 + y^2 - 4x + 6y$, find the critical point(s) and classify them.
4. Find the critical points of
$$f(x) = e^{-x^2}$$
and determine whether they are maxima or minima.
5. Given $f(x) = x^3 - 6x^2 + 9x + 2$, find the critical points and determine the intervals where f is increasing and decreasing.

Critical Points and Classification

Example 1: Find critical points of $f(x) = x^3 - 3x^2 + 4$.

$$f'(x) = 3x^2 - 6x = 3x(x-2) \implies x = 0, 2$$

Example 2: For $f(x) = x^4 - 4x^3 + 6x^2$,

$f'(x) = 4x^3 - 12x^2 + 12x = 4x(x^2 - 3x + 3)$, only real critical point $x = 0$,

$$f''(0) = 12 > 0 \implies x = 0 \text{ is local minimum.}$$

Critical Points and Classification

Example 3: $f(x,y) = x^2 + y^2 - 4x + 6y$ Critical point at $(2,-3)$ since

$$f_x = 2x - 4 = 0, \quad f_y = 2y + 6 = 0.$$

Example 4: For $f(x) = e^{-x^2}$, critical point at $x = 0$ since

$$f'(x) = -2xe^{-x^2} = 0 \Rightarrow x = 0.$$

$$f''(0) = -2 < 0 \implies x = 0 \text{ is local maximum.}$$

Example 5: $f(x) = x^3 - 6x^2 + 9x + 2$

$$f'(x) = 3(x-1)(x-3), \quad x = 1, 3 \text{ critical points.}$$

Increasing on $(-\infty, 1)$, decreasing on $(1,3)$, increasing on $(3,\infty)$. Local max at $x = 1$, local min at $x = 3$.

How to Find Global Maximum and Minimum of a Function

Step 1: Identify the domain
Step 2: Find critical points Compute $f'(x)$ and solve $f'(x) = 0$ to find candidates.
Step 3: Evaluate endpoints If domain is closed or bounded, evaluate f at endpoints.
Step 4: Determine global maxima/minima Compare function values at critical points and endpoints;

$$\max f(x) \quad \text{and} \quad \min f(x)$$

occur at these points.

Additional Notes: - Use second derivative test or first derivative test to classify critical points as local max/min. - For functions not bounded, global max or min may not exist.

Example: To find global max/min of $f(x) = x^3 - 3x^2 + 4$ on $[-1, 3]$, find critical points, evaluate f at critical points and at $-1, 3$, then compare values.

Problem Statement

Find the **global maximum** and **global minimum** of the function

$$f(x) = x^3 - 3x^2 + 4$$

on the closed interval $[-1, 3]$.

Approach:

1. Find critical points by setting $f'(x) = 0$.
2. Evaluate f at critical points within the interval.
3. Evaluate f at the endpoints -1 and 3.
4. Compare all values to determine global max and min.

Finding Critical Points

Compute the first derivative:

$$f'(x) = 3x^2 - 6x = 3x(x-2).$$

Set derivative equal to zero to find critical points:

$$3x(x-2) = 0 \implies x = 0 \quad \text{or} \quad x = 2.$$

Both points $x = 0$ and $x = 2$ lie inside the interval $[-1, 3]$, so consider both for evaluation.

Evaluating $f(x)$ at Critical Points and Endpoints

Compute values:

$$f(-1) = (-1)^3 - 3(-1)^2 + 4 = -1 - 3 + 4 = 0,$$

$$f(0) = 0^3 - 3(0)^2 + 4 = 4,$$

$$f(2) = 2^3 - 3(2)^2 + 4 = 8 - 12 + 4 = 0,$$

$$f(3) = 3^3 - 3(3)^2 + 4 = 27 - 27 + 4 = 4.$$

Values at points:

$$f(-1) = 0, \quad f(0) = 4, \quad f(2) = 0, \quad f(3) = 4.$$

Conclusion: Global Maximum and Minimum

From the evaluated values:

- The **global maximum** is 4 at $x = 0$ and $x = 3$.
- The **global minimum** is 0 at $x = -1$ and $x = 2$.

Summary: To find global extrema on a closed interval, evaluate the function at critical points within the interval and at the endpoints, then pick the highest and lowest values.

Find Global Max/Min of $f(x) = x^3 - 3x^2 + 4$

On the interval $[-1.5, 3.5]$, find the global maximum and minimum of

$$f(x) = x^3 - 3x^2 + 4.$$

Compute derivative:

$$f'(x) = 3x^2 - 6x = 3x(x - 2).$$

Set derivative to zero to find critical points:

$$x = 0, \quad x = 2.$$

Evaluate $f(x)$ at Critical Points and Endpoints

Calculate values:

$$f(-1.5) = -6.125, \quad f(0) = 4, \quad f(2) = 0, \quad f(3.5) = 10.125.$$

These correspond to function values at the endpoints and critical points within the interval.

Conclusion: Extrema of $f(x) = x^3 - 3x^2 + 4$ on $[-1.5, 3.5]$

- **Local maxima:** $f(0) = 4$ and $f(3.5) = 10.125$ are points where the function reaches local highs.
- **Local minimum:** $f(2) = 0$ is a local low point.
- **Global maximum:** The highest value on the interval is 10.125 at $x = 3.5$.
- **Global minimum:** The lowest value on the interval is -6.125 at $x = -1.5$.

Summary: To find global extrema, evaluate function values at local extrema and endpoints, then determine the absolute highest and lowest.

Definition of Affine Set

A set $C \subseteq \mathbb{R}^n$ is **affine** if for any two points $x_1, x_2 \in C$ and any scalar $\theta \in \mathbb{R}$, the point

$$\theta x_1 + (1-\theta)x_2 \in C$$

That is, C contains the entire **line** passing through any two points in it.

More generally, for points $x_1, \ldots, x_k \in C$, any **affine combination** of the form

$$\theta_1 x_1 + \cdots + \theta_k x_k, \quad \text{with } \sum_{i=1}^{k} \theta_i = 1$$

also lies in C.

Properties and Examples

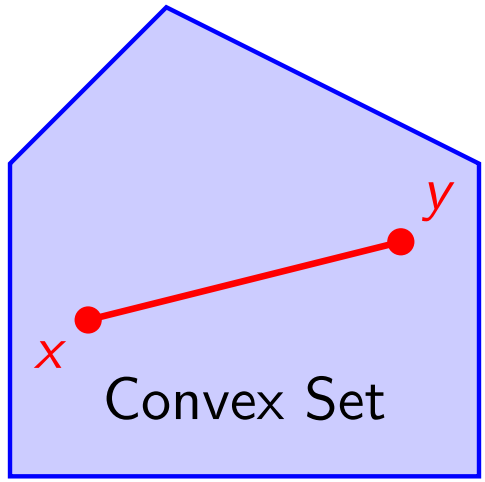
- An affine set contains every line through any pair of its points.
- Every affine set is a translation of a linear subspace.
- Examples include:
  - Lines and planes in $\mathbb{R}^n$
  - Hyperplanes (like a flat surface in 3D space)
  - Solutions sets of linear equations

Convex Set

Definition

A set C in a vector space is **convex** if for any two points $x, y \in C$, the line segment connecting them lies entirely within C. Formally,

$$\text{If } x, y \in C, \text{ then } \lambda x + (1-\lambda)y \in C \quad \forall \lambda \in [0,1].$$



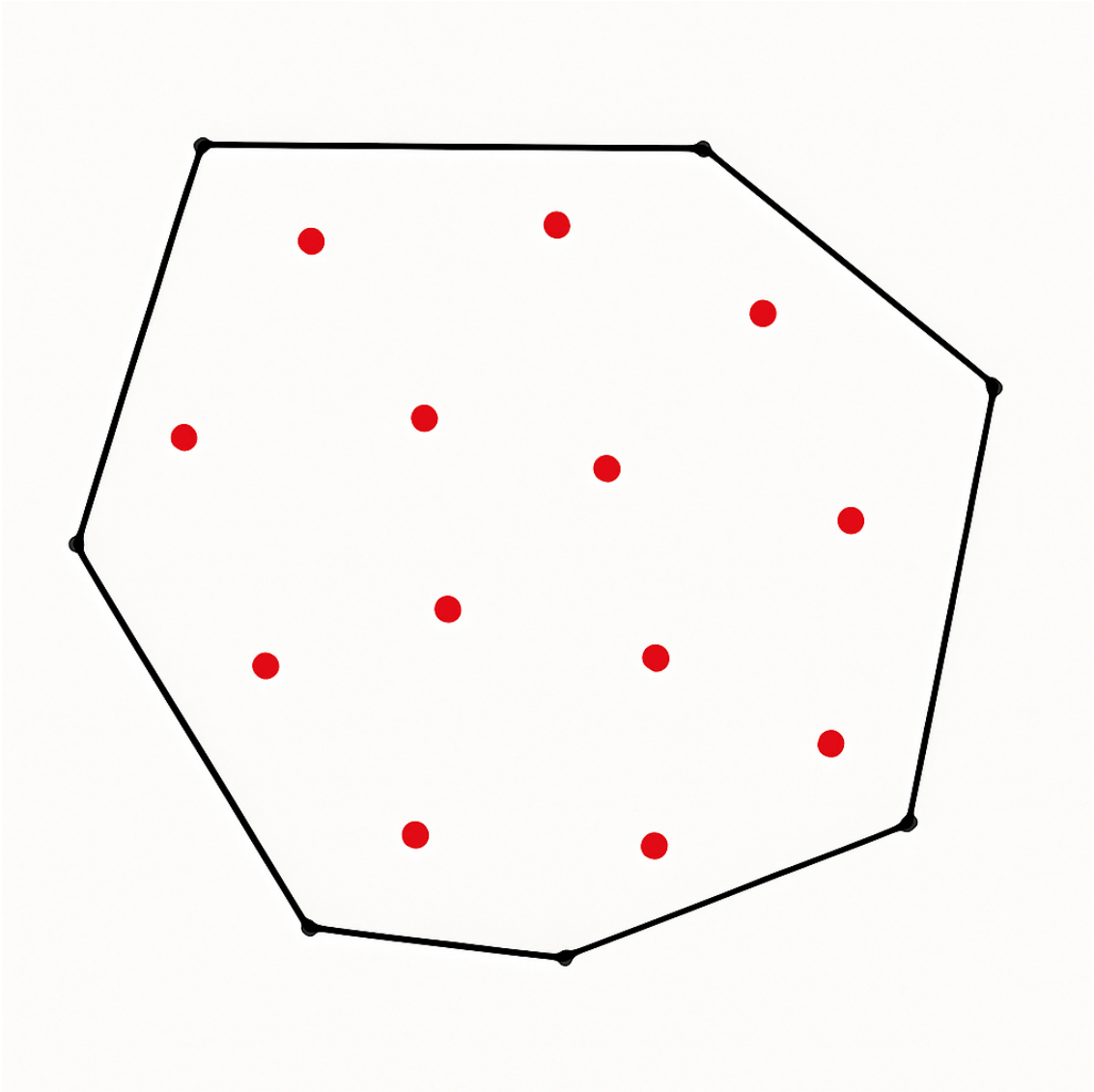
- Affine sets differ from convex sets in that they contain entire lines, not just line segments.

Convex Hull

- The **convex hull** of a set of points $S \subseteq \mathbb{R}^n$ is the **smallest convex set** containing S.
- Intuitively, imagine stretching a **rubber band** around the outermost points in S; when released, it snaps tightly around the points forming the convex hull.
- Formally, the convex hull is the set of all **convex combinations** of points in S:

$$\left\{ \sum_{i=1}^{k} \theta_i x_i \mid x_i \in S, \theta_i \geq 0, \sum_{i=1}^{k} \theta_i = 1, k \in \mathbb{N} \right\}$$

Convex Hull



Applications of Convex Hull

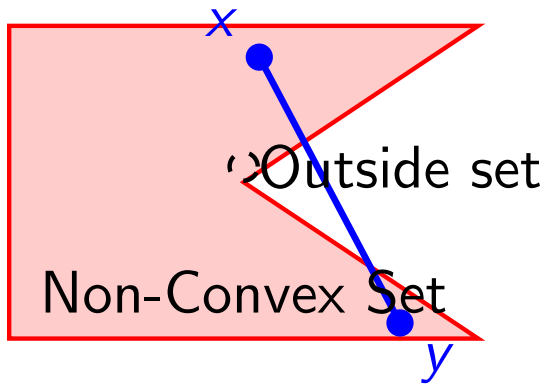
- Used in computational geometry algorithms
- Image processing: shape analysis and object recognition
- Robotics: collision detection and path planning
- Geographic information systems (GIS) to define bounding regions

Non-Convex Set

Definition

A set S is **non-convex** if there exist points $x, y \in S$ such that the line segment connecting them is **not entirely** contained in S. Formally,

$$\exists x, y \in S, \quad \exists \lambda \in (0,1) \text{ such that } \lambda x + (1-\lambda)y \notin S.$$

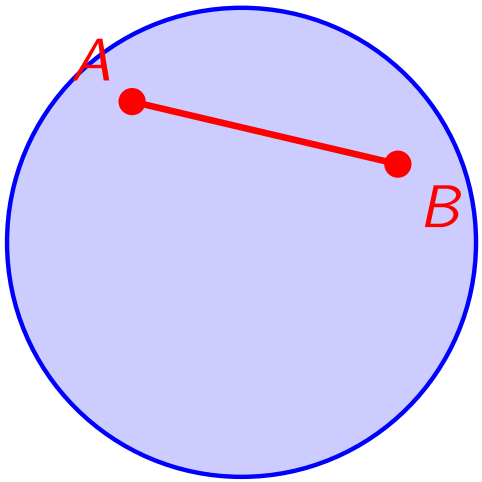


- The blue line segment crosses outside the shaded red region.
- This demonstrates the set is **non-convex**.

Example: Convex Set

Convex Set (Disk in $\mathbb{R}^2$)

The set $C = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 \leq 1\}$ is convex.



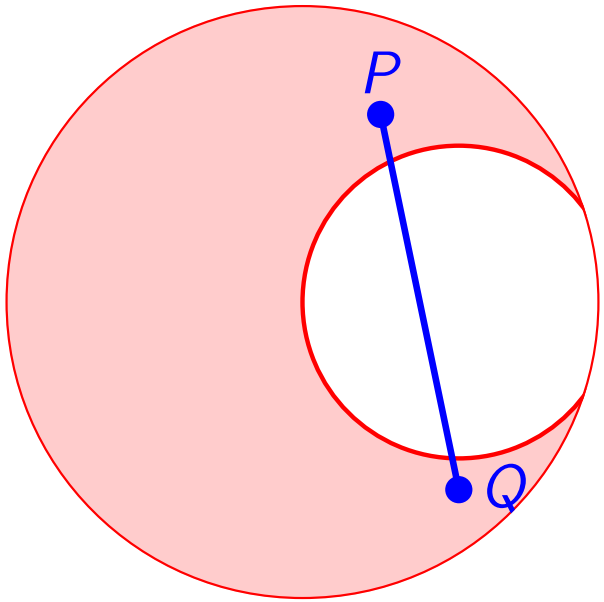
- The line segment between any two points within the disk stays entirely inside the blue region.

Example: Non-Convex Set

Non-Convex Set (Crescent Shape)

Let S be a "crescent" shape formed by subtracting a smaller disk from a bigger one:

$$S = \{(x, y) \mid x^2 + y^2 \leq 2^2\} \setminus \{(x, y) \mid (x-1)^2 + y^2 < 1^2\}$$



- The line segment from P to Q exits the region, illustrating non-convexity.

Hyperplanes in $\mathbb{R}^n$

- A **hyperplane** is a flat affine subspace of dimension $n-1$ in $\mathbb{R}^n$.
- It can be defined as the set of points $x \in \mathbb{R}^n$ satisfying a linear equation:

$$\{x \in \mathbb{R}^n \mid a^\top x = b\}$$

  where $a \in \mathbb{R}^n$ is a non-zero vector (normal vector), and $b \in \mathbb{R}$ is a scalar.
- Geometrically, a hyperplane divides the space into two halves.

Halfspaces

- A **halfspace** is one of the two parts into which a hyperplane divides $\mathbb{R}^n$.
- The two halfspaces defined by a hyperplane $a^\top x = b$ are:

$$H_+ = \{x \in \mathbb{R}^n \mid a^\top x \leq b\}, \quad H_- = \{x \in \mathbb{R}^n \mid a^\top x \geq b\}$$

- Halfspaces are convex sets.
- In optimization, halfspaces represent linear inequality constraints.

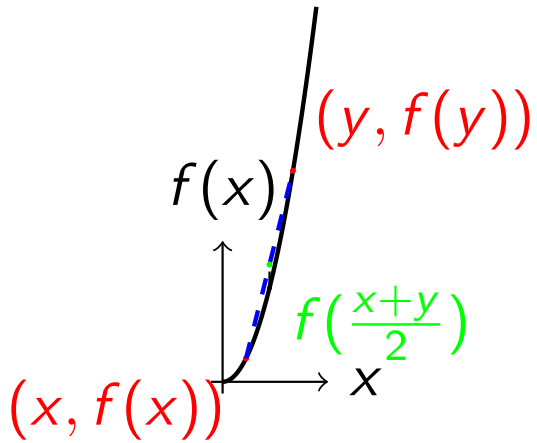
Convex Function

Definition

A function $f : \mathbb{R} \to \mathbb{R}$ is **convex** if for all x, y in its domain and for any $\lambda \in [0, 1]$:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

Intuitively, the graph of f lies below the straight line connecting any two points on it.



- The black point on the curve lies below the green point on the chord, illustrating convexity.

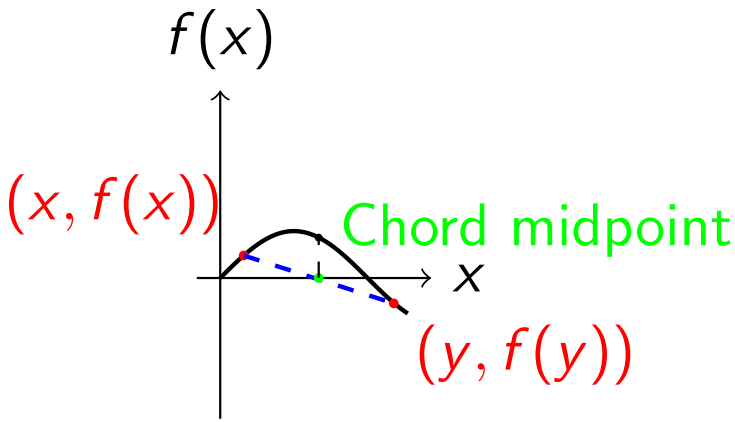
Non-Convex Function

Definition

A function f is **non-convex** if it violates the convexity condition for some x, y and $\lambda \in [0, 1]$, i.e.,

$$f(\lambda x + (1 - \lambda)y) > \lambda f(x) + (1 - \lambda)f(y).$$

Intuitively, the graph of f lies above the line segment joining some two points on it.



- The black point on the curve lies **above** the green point on the chord, illustrating non-convexity.

Methods to Check Convexity of a Function

- **Definition Method:** Check if for all x, y and $\lambda \in [0, 1]$:

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y).$$

- **First Derivative Test (for differentiable functions):** The function f is convex if and only if its domain is convex and

$$f(y) \geq f(x) + \nabla f(x)^T (y - x), \quad \forall x, y.$$

- **Second Derivative Test (for twice differentiable functions):** The function f is convex if and only if its Hessian matrix satisfies

$$\nabla^2 f(x) \succeq 0, \quad \text{(positive semidefinite)} \quad \forall x.$$

- **Graphical Check:** The graph of a convex function lies below the chord connecting any two points.

First Derivative Test for Convexity

Definition: A differentiable function f on interval I is convex if for all $x, y \in I$,

$$f(y) \geq f(x) + f'(x)(y - x).$$

This inequality means the function always lies **above** its tangent lines.

Geometric Interpretation:

- For any point x, the tangent line at x is a *supporting line* that touches but never dips below the curve.
- The slope $f'(x)$ is monotonically non-decreasing for convex functions.

Example: Applying the First Derivative Test

Check convexity of $f(x) = e^x$:

- Derivative:
$$f'(x) = e^x > 0 \quad \forall x \in \mathbb{R}.$$
- Tangent line at x:
$$L(y) = f(x) + f'(x)(y - x) = e^x + e^x(y - x) = e^x(1 + y - x).$$
- Check inequality:
$$e^y \overset{?}{\geq} e^x(1 + y - x).$$
- **Conclusion:** $f(x) = e^x$ is convex on $\mathbb{R}$.

Second Derivative Test for Convexity

For twice differentiable functions, the second derivative test states:

$$f \text{ is convex} \iff \nabla^2 f(x) \geq 0 \quad \forall x$$

For functions of one variable:

$$f \text{ is convex} \iff f''(x) \geq 0 \quad \forall x$$

This means the function's curvature is always upward or flat.

Example: Check $f(x) = e^x$ Using Second Derivative

Step 1: Compute first derivative:

$$f'(x) = e^x$$

Step 2: Compute second derivative:

$$f''(x) = e^x$$

Step 3: Since

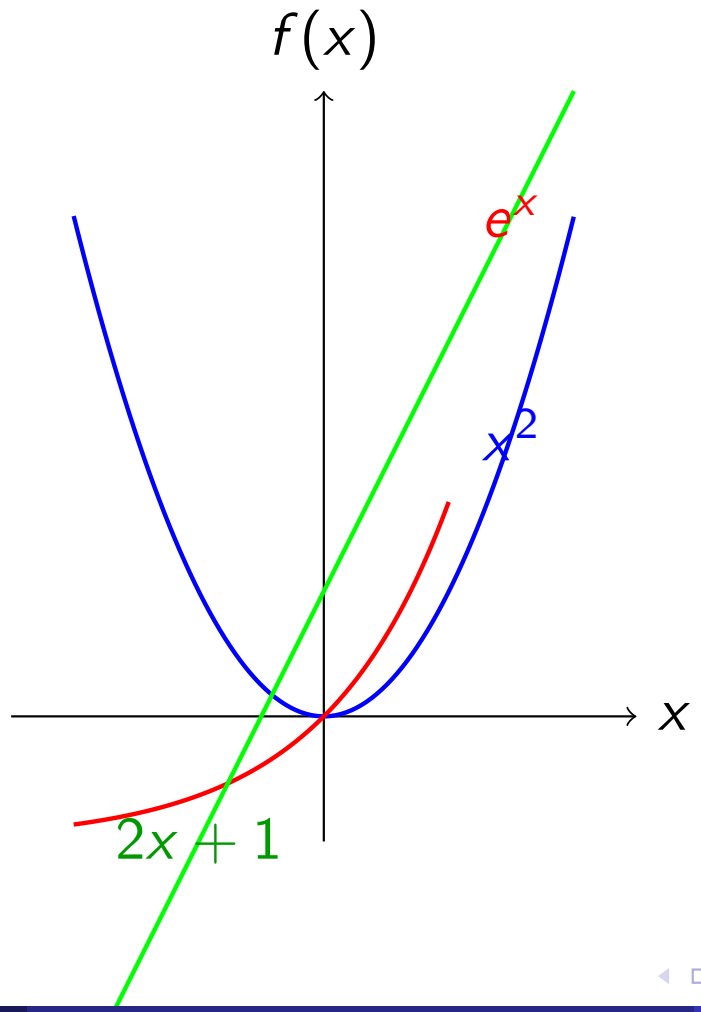
$$f''(x) = e^x > 0 \quad \forall x \in \mathbb{R},$$

the function is convex.

Conclusion: The exponential function $f(x) = e^x$ is convex on $\mathbb{R}$ as its second derivative is always positive.

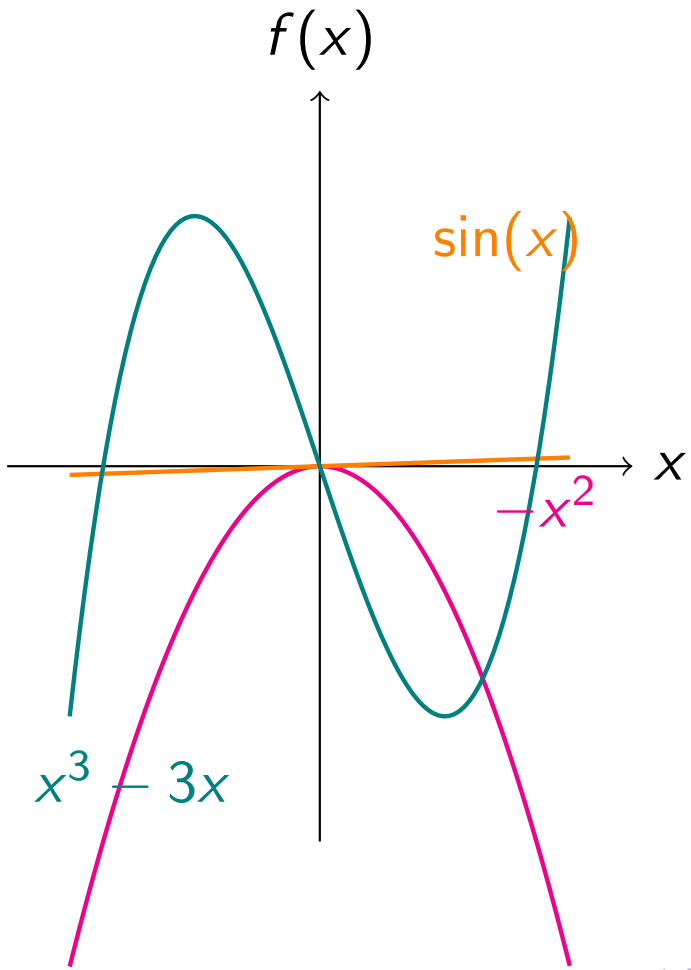
Examples of Convex Functions

- $f(x) = x^2$ (Upward-opening parabola)
- $f(x) = e^x$ (Exponential function)
- $f(x) = 2x + 1$ (Straight line)



Examples of Non-Convex Functions

- $f(x) = -x^2$ (Downward parabola)
- $f(x) = \sin(x)$ (Sine wave)
- $f(x) = x^3 - 3x$ (Cubic function)



Importance of Convex Functions in Optimization

- Convex functions **guarantee global optimality** because any local minimum is also a global minimum.
- This property **eliminates local traps**, making optimization problems easier to solve.
- The well-defined curvature allows **efficient algorithms** such as gradient descent and Newton's method to converge quickly.
- Convex optimization problems are **mathematically tractable** and widely used in machine learning, economics, and engineering.

Advantages of Convexity in Optimization Algorithms

- **Strong duality** holds for convex problems, enabling powerful dual optimization techniques.
- **Smoothness** and **Lipschitz continuity** of gradients often hold, improving convergence rates.
- Convex problems admit **polynomial-time algorithms**, ensuring scalability.
- Robustness to noise and approximate computations due to convex structure.